

An Exact Approach to Solving the Drop the Handkerchief Game

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Abstract

Drop the Handkerchief (DTH) is a death game played during the final arc "Surpassing The Leader" of the manga *Usogui* by Toshio Sako. It is a fully observed, two-player, simultaneous action, zero-sum stochastic game. Owing to its deadly nature, we solve the complete 60-action timing model under the frozen survival law of Section 2. Once a player can no longer survive an injection, the game never reads their cumulative time dead (TTD) again. Quotienting by that observation collapses the 5.27×10^9 -state transition-closed domain down to 289,374,121 equivalence classes (a factor of 18.2). A potential function organizes those classes into 1,201 layers and permits one backward sweep over the finite DAG. Every local value is accepted with a mixed-strategy gap of at most 10^{-6} against its full 60×60 stage matrix. At the initial state, $V = 0.08985 \pm 0.00061$, so the first Dropper wins with probability 0.54493 ± 0.00031 under optimal play. No unilateral change to the selected policy can improve win probability by more than 6.005×10^{-4} . The largest continuous run covered 98.1% of the table in 5,432 seconds on one twelve-core desktop.

1 Introduction

DTH is the pure timing core of the Surpassing the Leader arc wager in *Usogui*. Two players alternate as *Dropper* and *Checker*. Each round, both choose a second in $\{1, \dots, 60\}$. A successful check adds the inclusive delay to the Checker's squandered time (ST), then swaps the roles. A failed check injects that ST plus sixty seconds. Revival follows the fixed law in Figure 1; otherwise the Checker loses.

The game is finite but large, and each round may require randomization. The solution follows four steps:

1. quotient profiles once failure is certainly fatal (Section 4);
2. rank every continuing transition with an increasing potential (Section 5);
3. solve each 60×60 stage game and retain a gap certificate (Section 6);
4. use the matrix's 61 transition values to avoid scanning all 3,600 cells when testing pure play (Section 3).

Section 7 reports the computation and its result. Section 8 states what is and is not established.

*Code and instructions to reproduce the results are available in the project repository.

The key game theory object

A state keeps exactly the past that can still affect play. At a state x , each player chooses a probability distribution over the sixty seconds:

$$\Delta_{60} = \left\{ \sigma \in \mathbb{R}^{60} : \sigma_i \geq 0, \sum_{i=1}^{60} \sigma_i = 1 \right\}.$$

These distributions are mixed strategies. The stage matrix $M(x)$ gives the payoff for each action pair, so

$$V(x) = \max_{\sigma_d \in \Delta_{60}} \min_{\sigma_c \in \Delta_{60}} \sigma_d^\top M(x) \sigma_c.$$

The matrix entries contain either a terminal payoff or the value of a later state. This is Bellman recursion. Once transitions are ordered as a DAG, later values can be substituted backward. Finally, $V = P(\text{win}) - P(\text{loss}) = 2P(\text{win}) - 1$, hence $P(\text{win}) = (1 + V)/2$.

2 The game

Definition 2.1 (State). *A player's profile is $z = (s, t)$: their current ST and TTD. A live state is the ordered pair $x = (z_c, z_d) = (s_c, t_c, s_d, t_d)$, written from the perspective of the current Dropper. Here $s_c, s_d \in \{0, \dots, 299\}$ and $t_c, t_d \in \{0, \dots, 300\}$. Terminal payoffs are +1 when the current Dropper wins and -1 when they lose.*

Definition 2.2 (Actions and timing). *Both players simultaneously choose seconds $d, c \in \{1, \dots, 60\}$ (drop and check). The check succeeds iff $d \leq c$, and then the Checker's ST grows by the inclusive interval $\ell = c - d + 1 \in \{1, \dots, 60\}$; equal seconds cost one second, not zero. If $s_c + \ell \geq 300$ the cylinder overflows and the checker dies: the transition ends with payoff +1 for the Dropper. Overflow is uniformly fatal: a cylinder at capacity leaves no revival headroom for any dose. Otherwise play continues from*

$$(s_c, t_c, s_d, t_d) \xrightarrow{\text{success, lag } \ell} (s_d, t_d, s_c + \ell, t_c),$$

with the roles swapped (the previous Checker drops next).

For example, $d = 12$ and $c = 15$ gives a successful check with $\ell = 15 - 12 + 1 = 4$. If instead $c = 10$, the check fails and the Checker receives their accumulated ST plus sixty seconds.

Definition 2.3 (Failure, dose, and revival). *On a failed check ($d > c$) the checker is injected with the dose $q = s_c + 60$. Survival is possible iff*

$$q < 300 \wedge q + t_c \leq 300 \iff s_c \leq 239 \wedge s_c + t_c \leq 240, \quad (1)$$

Equality at $q + t_c = 300$ remains revival-eligible. A profile is revival-eligible when (1) holds and failure-fatal otherwise. These terms describe the next failed check; both players are still alive in a live state. When the profile is revival-eligible, the checker revives with probability

$$p(s_c, t_c) = 0.95 \left(1 - \frac{s_c}{240} \right) 0.75^{t_c/60}, \quad (2)$$

their cylinder resets, and their TTD grows by the dose:

$$(s_c, t_c, s_d, t_d) \xrightarrow{\text{failure, revive}} (s_d, t_d, 0, t_c + s_c + 60) \quad \text{with probability } p(s_c, t_c),$$

else the transition ends with payoff +1. When (1) fails, the injection is fatal with certainty and $p(s_c, t_c) \equiv 0$.

Frozen revival probability over ST and TTD

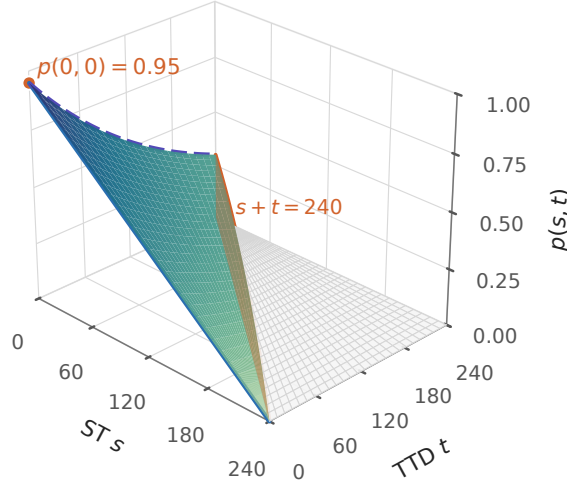


Figure 1: The frozen revival law (2). Color and height both encode probability. The blue edge isolates the linear loss from ST; the purple edge isolates the exponential loss from TTD. The pale plateau is the zero-probability region containing (240, 240). The orange curtain marks $s + t = 240$: probability is positive on the curtain, except at its $s = 240$ endpoint, and drops to zero immediately beyond it.

The source gives no numerical revival odds, but it fixes the qualitative geometry: a failed check injects the vial contents plus sixty seconds against a five-minute cap and cumulative deaths/TTD lower revival prospects enough to change strategy [1]. These facts motivate the separable form in (2). Since $q = s + 60$, the acute factor $1 - s/240 = (300 - q)/240$ is the simplest monotone path from a bare injection to the lethal dose, so each second of ST consumes an equal share of acute headroom. TTD records damage carried through earlier revivals; $0.75^{t/60}$ makes each additional death-minute retain a fixed fraction of the chance remaining after the acute dose. The source supports the signs and endpoint, not linearity, multiplicative separation, or the constants 0.95 and 0.75; those are explicit modeling choices.

A deliberately coarse sensitivity check varies the baseline alone. The six-action abstraction measures time in ten-second units, so actions $1, \dots, 6$ represent seconds $10, 20, \dots, 60$. Four exact solves at baselines 0.85, 0.90, 0.95, and 0.99 place the root value between 0.09639 and 0.09921, a span of 0.0028, while the Dropper’s first-action mass moves from 0.387 to 0.257. The value is therefore fairly stable to this one parameter at coarse resolution, but the policy is not. This check does not calibrate the TTD factor 0.75, test multiplicative separation, or establish full-resolution robustness.

Values are from the current Dropper’s perspective and negate when the roles swap.

3 Finiteness and stage structure

Lemma 3.1 (Damage strictly increases). *Let $R(x) = s_c + t_c + s_d + t_d$. Every live transition strictly increases R : a successful lag ℓ adds $\ell \geq 1$, and a revived failure adds exactly 60 (the dose $s_c + 60$ enters TTD while the ST s_c resets).*

Corollary 3.2 (The game is finite). $R \leq 299 + 300 + 299 + 300 = 1198$ on live states, so every play reaches a terminal outcome within 1199 transitions. The game therefore has a unique value $V(x) \in [-1, 1]$ at every live state, obtained by backward induction over the finite directed acyclic transition graph [2], solving at each state the zero-sum simultaneous stage game by von Neumann's minimax theorem [3]; the construction is the finite, absorbing case of a Shapley stochastic game [4].

Mixed strategies matter even in a two-action game. In

$$\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

either fixed row can be punished by a column giving -1 , while choosing each row with probability $1/2$ guarantees 0 . DTH uses the same idea with sixty actions and state-dependent entries.

Proposition 3.3 (The 61 transition classes). *Fix a live state x and let V assign values to its children. Write the Dropper's continuation payoff after a success of lag ℓ or a failure as*

$$S_\ell = \begin{cases} +1, & s_c + \ell \geq 300, \\ -V(s_d, t_d, s_c + \ell, t_c), & \text{else,} \end{cases}$$

$$F = \begin{cases} +1, & (s_c, t_c) \text{ failure-fatal,} \\ p(-V(s_d, t_d, 0, t_c + s_c + 60)) + (1 - p), & \text{else.} \end{cases}$$

Then the 60×60 stage matrix of x , with the Dropper as row player, is

$$M(x)_{d,c} = \begin{cases} S_{c-d+1}, & c \geq d, \\ F, & c < d, \end{cases} \quad (3)$$

and $V(x) = \text{val}(M(x))$. In particular $M(x)$ carries at most 61 distinct entries: it is Toeplitz on and above the diagonal and constant below it.

Proof. A successful cell depends on the actions only through the lag $\ell = c - d + 1$, and every failed cell has the identical transition distribution; the entries are the expected continuation values, negated for the role swap. The Bellman equation is Corollary 3.2. $\square$

Lemma 3.4 ($O(60)$ saddle reductions). *Row d of (3) contains exactly the prefix $S_1, \dots, S_{61-d}$ together with F when $d > 1$, and column c contains the prefix $S_1, \dots, S_c$ together with F when $c < 60$. Hence with prefix scans $m_j = \min_{i \leq j} S_i$ and $M_j = \max_{i \leq j} S_i$,*

$$\max_d \min_c M(x)_{d,c} = \max_d \begin{cases} m_{60}, & d = 1, \\ \min(m_{61-d}, F), & d > 1, \end{cases}$$

$$\min_c \max_d M(x)_{d,c} = \min_c \begin{cases} \max(M_c, F), & c < 60, \\ M_{60}, & c = 60, \end{cases}$$

computable in $O(60)$ operations, exactly (min and max select one of their arguments, so the scans introduce no rounding), from the 61 class values alone.

Figure 2 draws the layout. Each row contains a prefix of the S array, never an arbitrary subset. Two linear scans therefore reproduce every row minimum and column maximum without materializing the matrix.

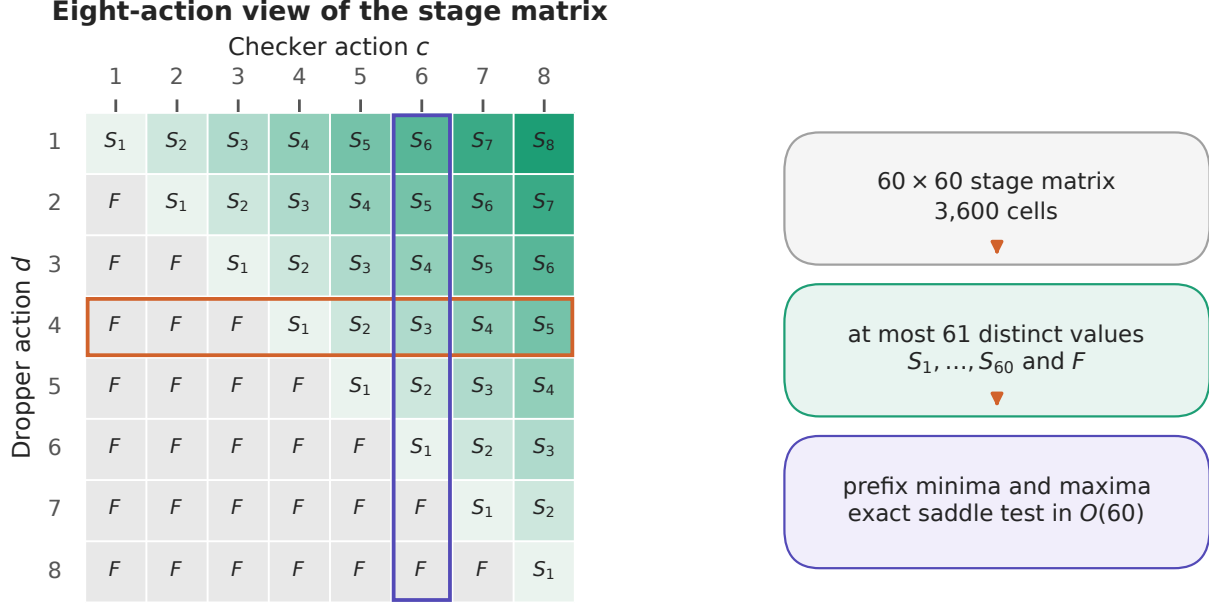


Figure 2: An 8×8 view of the stage matrix (3). The lag $\ell = c - d + 1$ is constant on diagonals, while every failed check has value F . Row minima and column maxima reduce to prefix scans (Lemma 3.4), giving the same result as a scan of all 3,600 cells.

4 The failure-fatal quotient

The model reads a player's TTD only in the revival probability (2). Once failure is certainly fatal, the exact TTD has no remaining effect.

For example, profiles (250, 60) and (250, 300) both receive dose 310 on a failed check, so failure is fatal in either case. Success only raises ST and cannot restore revival. Their TTDs can therefore share the sentinel (250, $\perp$).

Lemma 4.1 (Failure-fatality is absorbing). *If the profile (s, t) is failure-fatal, then (s', t) is failure-fatal for every $s' \geq s$. Since a successful check raises a player's ST and leaves their TTD unchanged, and a failed check either ends the game for them or requires (1), no continuing transition can make the profile revival-eligible again.*

Theorem 4.2 (Quotient soundness). *Define $\varphi(s, t) = (s, t)$ when (s, t) is revival-eligible and $\varphi(s, t) = (s, \perp)$ otherwise. Apply this map to each player's profile independently. Then any two live states with equal images under φ have identical stage matrices up to φ -identification of their children, and hence identical values: V factors through φ .*

Proof. Compare the 61 classes of Proposition 3.3 at two states that differ only in a failure-fatal player's TTD. Success classes never read TTD, and the child's φ -image is determined by the parent's φ -image and ℓ : a revival-eligible profile retains its TTD, which fixes its child's classification, and a failure-fatal profile stays so by Lemma 4.1. So success classes map to φ -identified children. The failure class reads the checker's TTD only through p : if the checker profile is failure-fatal, $p \equiv 0$ and $F = +1$ with no child; if revival-eligible, the TTD is not quotiented and the value is unchanged. Induction over the finite acyclic graph (Corollary 3.2) closes the argument. $\square$

Proposition 4.3 (Counting). *Over the transition-closed TTD domain $\{0\} \cup [60, 300]$ (Lemma 4.4), one player has $300 \times 242 = 72,600$ profiles. The quotient keeps all 16,711 revival-eligible profiles distinct and replaces the failure-fatal TTD values at each ST with one sentinel. This leaves $16,711 + 300 = 17,011$ classes per player, and therefore*

$$17,011^2 = 289,374,121$$

two-player classes, compared with $72,600^2 = 5,270,760,000$ before quotienting.

Lemma 4.4 (Domain closure). *The initial state has both TTDs in $\{0\} \cup [60, 300]$, and the domain is transition closed: revival requires $s_c + t_c \leq 240$, so a failure child’s TTD $t_c + s_c + 60$ lies in $[60, 300]$, and success preserves TTDs. A revival-eligible profile with TTD in 1..59 is thus unreachable from the domain; the implementation rejects states containing one. A failure-fatal TTD is discarded by φ exactly.*

Figure 3 shows why the reduction is large. Revival-eligible profiles make up 16,711, or 23%, of the one-player domain. For two players, both profiles are revival-eligible in only 5.3% of states. We checked the survival condition, absorption, and these counts by exhaustive enumeration over all 72,600 one-player profiles.

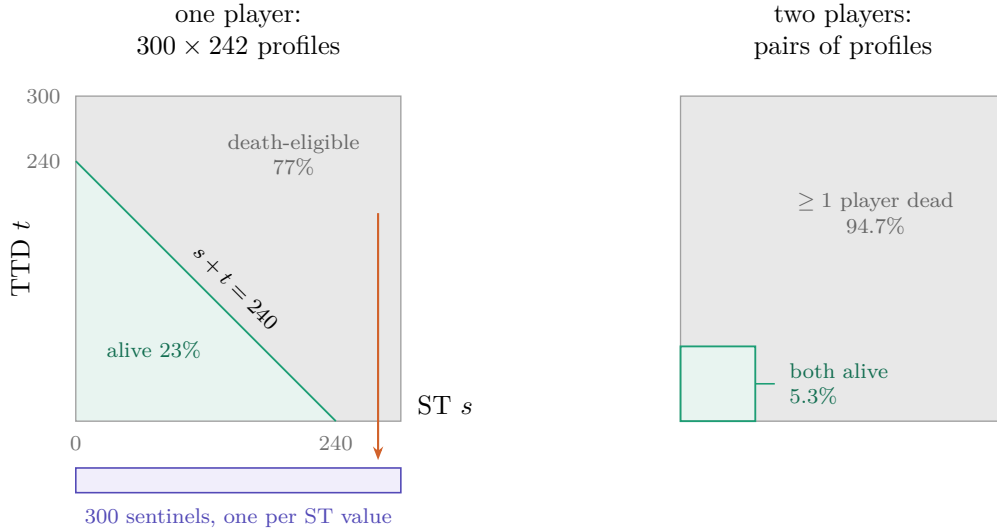


Figure 3: The quotient for one player (left) and for pairs of players (right). Each vertical fibre in the death-eligible region collapses to one sentinel per ST value. The left panel is schematic because TTD values 1, . . . , 59 are unreachable (Lemma 4.4); the percentages report exact discrete counts on the transition-closed domain. The compact labels “death-eligible” and “dead” mean failure-fatal; neither means that a player has already left the game.

5 The potential and the sweep schedule

The potential is not payoff, value, or strategic advantage. It is only a progress score used to order the states.

Definition 5.1 (Potential). *For a profile let $\rho(s, t) = t$ if (s, t) is revival-eligible and $\rho(s, t) = 301$ otherwise; for a class let*

$$\Phi(x) = s_c + s_d + \rho(s_c, t_c) + \rho(s_d, t_d) \in \{0, \dots, 1200\}.$$

Theorem 5.2 (Strict increase). *Every live transition strictly increases Φ . By cases on the moving profile:*

- *successful lag ℓ , the profile stays on the same side of the revival boundary: $\Delta\Phi = \ell \geq 1$;*
- *successful lag ℓ , the profile becomes failure-fatal: $\Delta\Phi = \ell + 301 - t \geq \ell + 61$, since revival-eligibility implies $t \leq 240$;*
- *failure, revived profile remains revival-eligible: $\Delta\Phi = 60$ (the ST resets to zero while TTD absorbs the dose);*
- *failure, revived profile becomes failure-fatal: $\Delta\Phi = 301 - (s_c + t_c) \geq 61$ by (1).*

Corollary 5.3 (Layered backward induction). *No transition stays within a Φ -layer, so a descending sweep solves every child before its parents. Any layer- P residual work need only complete before layer $P - 1$ begins. The 289,374,121 classes occupy all 1,201 layers; the largest, $\Phi = 374$, holds 1,678,715 classes. Define the one-player bucket*

$$B_a = \{(s, t) \text{ quotient profiles} : s + \rho(s, t) = a\}.$$

Then layer P is exactly

$$\Phi^{-1}(P) = \bigcup_{\substack{0 \leq a \leq 600 \\ 0 \leq P-a \leq 600}} B_a \times B_{P-a}.$$

The quotient classes therefore form the layered DAG in Figure 4.

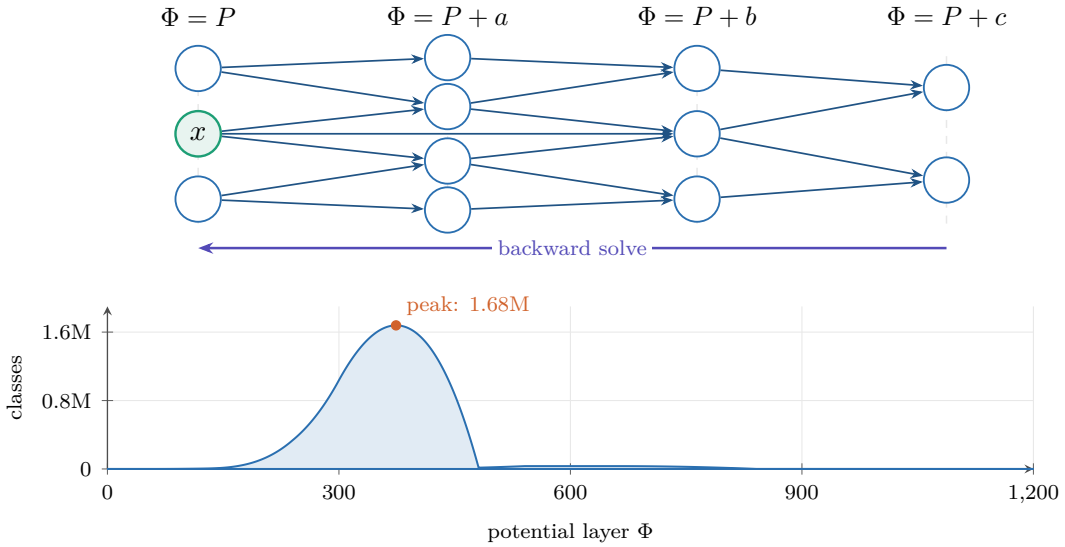


Figure 4: Top: representative transitions in the potential-ranked DAG. Bottom: the exact linear-scale width of all 1,201 layers. Every edge points right; the computation runs from right to left.

The strict increase was checked over all 1,035,541 live profile transitions. The schedule needs no reachability analysis, frontier queue, or transposition table; it simply solves unreachable classes as well.

6 Certified equilibria

Definition 6.1 (Certificate). *For a stage matrix M and mixed strategies σ_d (Dropper) and σ_c (Checker), let*

$$L = \min_c(\sigma_d^\top M)_c, \quad U = \max_d(M\sigma_c)_d, \quad \text{gap}_M(\sigma_d, \sigma_c) = \max(0, U - L).$$

The pair certifies the value $\hat{v} = (L + U)/2$ at tolerance τ_0 when $\text{gap}_M(\sigma_d, \sigma_c) \leq \tau_0$. Throughout, $\tau_0 = 10^{-6}$, and the gap is always measured against the full matrix (3).

For example, $L = 0.0898$ and $U = 0.0899$ place the matrix value inside that interval. The gap is 0.0001; it measures how much room remains between the two players' guarantees.

Proposition 6.2 (Certified midpoint). *$L \leq \text{val}(M) \leq U$, so a certificate at tolerance τ_0 pins $|\hat{v} - \text{val}(M)| \leq \tau_0/2$.*

Proof. σ_d guarantees the row player at least L against every reply, and σ_c caps the row player at U ; $\text{val}(M)$ sits between the two levels. $\square$

Lemma 6.3 (Matrix-value Lipschitz bound). *If $\max_{d,c} |M_{d,c} - \widehat{M}_{d,c}| \leq \epsilon$, then $|\text{val}(M) - \text{val}(\widehat{M})| \leq \epsilon$, since $|\sigma_d^\top (M - \widehat{M}) \sigma_c| \leq \epsilon$ for all mixed pairs.*

6.1 The solver ladder

The support of a mixed strategy is the set of actions it uses with positive probability:

$$\text{supp}(\sigma) = \{i : \sigma_i > 0\}.$$

Each class then follows the same three-step ladder.

1. **Pure saddle.** Lemma 3.4 computes the maximin and minimax in $O(60)$. If they meet, a deterministic action pair suffices.
2. **Equalizer.** A support guess reduces the mixed game to one linear system. A nearby solved support is tried first, then all sixty actions. Every candidate is checked against the full matrix.
3. **Linear programming.** The remaining matrices use the standard zero-sum primal and dual programs, followed by the same full-matrix check.

The equalizer systems, support rule, and LPs are stated in Appendix A.

6.2 Propagated error

The table stores certified midpoints, so a parent's matrix is assembled from child values that are themselves within a bounded distance of truth.

Proposition 6.4 (A priori accumulated bound). *Let every stored value carry a certificate at tolerance τ_0 and let $\delta(x)$ bound the longest live path from x to a terminal. Then the stored value $\widehat{V}(x)$ satisfies*

$$|\widehat{V}(x) - V(x)| \leq \delta(x) \frac{\tau_0}{2} \leq \frac{1201 - \Phi(x)}{2} \tau_0.$$

In particular the initial state obeys $|\widehat{V} - V|(0, 0, 0, 0) \leq 6.1 \times 10^{-4}$.

Proof. Induction up the DAG. Terminal entries are exact. If every child satisfies the bound with $\delta(x) - 1$, the assembled matrix is entrywise within $(\delta(x) - 1)\tau_0/2$ of the true one (expectations and negation are non-expansive), Lemma 6.3 transfers that to the value, and Proposition 6.2 adds the state's own $\tau_0/2$. Each live-to-live transition raises Φ by at least one (Theorem 5.2), so a play from x makes at most $1200 - \Phi(x)$ of them plus one terminal transition, bounding δ . $\square$

6.3 Policy certification

At each live class x , let $(\hat{\sigma}_d(x), \hat{\sigma}_c(x))$ be the accepted mixed pair. Using that pair whenever x is reached defines a behavioral policy $\hat{\pi}$. A deviation may replace every remaining choice, not only the next action.

Definition 6.5 (Policy security values). *For two complete policies, extend the value notation by*

$$V_x(\pi_d, \pi_c) = \mathbb{E}[\text{terminal payoff} \mid x, \pi_d, \pi_c].$$

The subscripts d, c identify the players by their roles at x ; both policies include play after later role swaps. The selected policies have security values

$$\begin{aligned} \underline{V}_{\hat{\pi}}(x) &= \min_{\pi_c} V_x(\hat{\pi}_d, \pi_c), \\ \overline{V}_{\hat{\pi}}(x) &= \max_{\pi_d} V_x(\pi_d, \hat{\pi}_c). \end{aligned}$$

The policy's own payoff lies between these values:

$$\underline{V}_{\hat{\pi}}(x) \leq V_x(\hat{\pi}_d, \hat{\pi}_c) \leq \overline{V}_{\hat{\pi}}(x).$$

Hence either player's gain from a unilateral deviation is bounded by $\overline{V}_{\hat{\pi}}(x) - \underline{V}_{\hat{\pi}}(x)$.

Let $\varepsilon(x)$ be the largest sum of accepted matrix gaps along any continuation from x to a terminal state. For a live state x , let y be a live child of x . Then

$$\varepsilon(x) = \begin{cases} 0, & x \text{ terminal,} \\ (U_x - L_x) + \max_y \varepsilon(y), & x \text{ live.} \end{cases} \quad (4)$$

The maximum is zero when x has no live child.

Proposition 6.6 (Full-policy bound). *The selected policy satisfies*

$$\begin{aligned} \underline{V}_{\hat{\pi}}(x) &\geq \hat{V}(x) - \frac{1}{2}\varepsilon(x), \\ \overline{V}_{\hat{\pi}}(x) &\leq \hat{V}(x) + \frac{1}{2}\varepsilon(x). \end{aligned}$$

Consequently,

$$\overline{V}_{\hat{\pi}}(x) - \underline{V}_{\hat{\pi}}(x) \leq \varepsilon(x) \leq (1201 - \Phi(x))\tau_0.$$

At the initial state, the payoff gap is at most 1.201×10^{-3} , so no unilateral deviation can improve win probability by more than 6.005×10^{-4} .

Proof. At a terminal, $\underline{V}_{\hat{\pi}} = \overline{V}_{\hat{\pi}} = \hat{V}$ and $\varepsilon = 0$. Suppose the two bounds hold for every live child of x . Negation at a role swap and averaging after revival preserve the largest child bound. Therefore

$$\begin{aligned}\underline{V}_{\hat{\pi}}(x) &\geq L_x - \frac{1}{2} \max_y \varepsilon(y), \\ \overline{V}_{\hat{\pi}}(x) &\leq U_x + \frac{1}{2} \max_y \varepsilon(y).\end{aligned}$$

Since $\hat{V}(x) = (L_x + U_x)/2$, recurrence (4) gives the two stated bounds.

Finally, $U_x - L_x \leq \tau_0$ at every live state, and every continuing transition raises Φ by at least one. Hence no path from x contains more than $1201 - \Phi(x)$ live states. $\square$

7 The computation

7.1 Sweep schedule

The potential turns the computation into a fixed order rather than a search. The solver starts at $\Phi = 1200$ and moves left across Figure 4 to $\Phi = 0$; within a layer, its bucket rectangles are independent and can run in parallel. Revival probabilities are evaluated once, and each completed layer is checkpointed before the next begins.

7.2 Results

The sweep covered all 289,374,121 classes across checkpointed sessions on a single twelve-core consumer desktop (AMD Ryzen 9 3900X). The final session — 283,764,990 classes, 98.1% of the table — completed in 5,432 seconds (1.51 hours); the remaining 5,609,131 classes were built in earlier, untimed sessions. The previously recorded single-core projection for the unquotiented, search-based design was 3.7–5.1 years; that ratio mixes three factors: the $18.2\times$ quotient, the sweep-for-search redesign, and twelve-way parallelism.

The solver routes show which optimizations actually mattered:

rung	classes	share
pure saddle (Lemma 3.4)	334,588	0.116%
equalizer, warm-started support	38,138	0.013%
equalizer, full support ($k = 60$)	288,810,400	99.805%
linear programming	190,995	0.066%

The full-support equalizer did almost all of the work. The warm solver rarely helped because its twelve-action cap could not represent the mostly full-support equilibria; linear programming was still needed for the final 0.066% of classes.

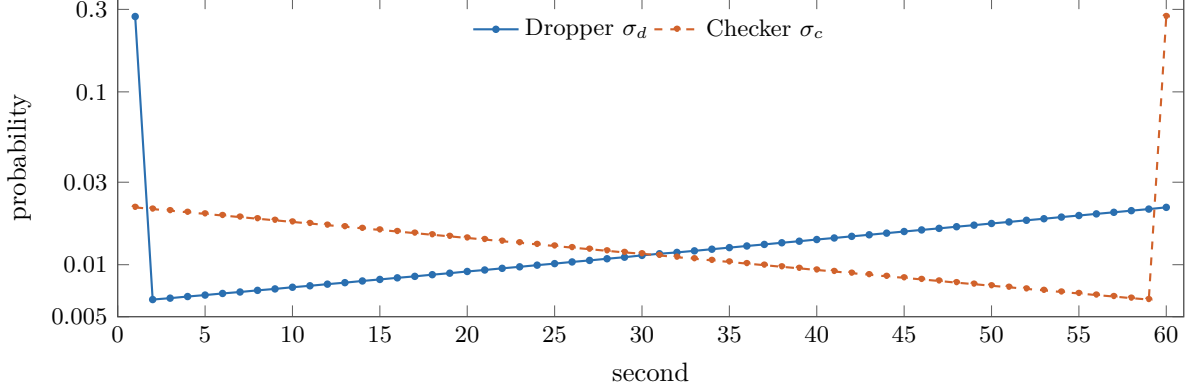


Figure 5: The first-round strategies of the selected full-game policy. Their local gap is 3.2×10^{-15} ; Proposition 6.6 shows how later gaps accumulate. Both strategies have full support.

The Python and Rust sweeps agreed byte-for-byte on their overlapping regions; 4,792 sampled Bellman re-solves differed by at most 4.84×10^{-7} . The error bound itself is Proposition 6.4.

Theorem 7.1 (Main result). *The complete 60-action timing model of Section 2 has a certified numerical solution on the transition-closed quotient domain of Lemma 4.4. Every one of its 289,374,121 classes carries a stored midpoint with local gap at most 10^{-6} . At the initial state,*

$$V(0, 0, 0, 0) = 0.08985 \pm 0.00061, \quad P(\text{first Dropper wins}) = 0.54493 \pm 0.00031.$$

For the selected full-game policy,

$$\overline{V}_{\hat{\pi}}(0, 0, 0, 0) - \underline{V}_{\hat{\pi}}(0, 0, 0, 0) \leq 1.201 \times 10^{-3}.$$

8 Scope of the claims

- **Rules-relative.** The results concern Section 2 and the frozen revival surface (2), not the leap-aware game with its asymmetric 61st second.
- **Certified numerics.** Values are round-to-nearest float64 midpoints with saddle-gap certificates, not rational or interval arithmetic. The structural theorems are exact; survival, absorption, counting, and potential increase were exhaustively enumerated.
- **Domain.** The table covers the transition-closed quotient domain; states containing a revival-eligible profile with TTD in 1.59 are outside it and are rejected, not approximated (Lemma 4.4).
- **Policies.** The table stores values and routing; accepted strategies are re-derived from child values. Figure 5 shows the first pair.

A Computational details

A.1 Sweep

One-player quotient profiles are indexed by $p \in \{0, \dots, 17,010\}$; a state class is stored at $17,011p_c + p_d$. For $P = 1200, 1199, \dots, 0$, the solver:

1. enumerates the rectangles $B_a \times B_{P-a}$;
2. builds $S_1, \dots, S_{60}$ and F from already solved children;
3. tries the pure, equalizer, and LP rungs in order;
4. recomputes L and U against the full matrix and stores their midpoint only when $U - L \leq \tau_0$;
5. checkpoints the completed layer before moving to $P - 1$.

Revival probabilities are evaluated once. Classes within one layer are independent.

A.2 Equalizer and LP fallbacks

Warm supports come from $(T_1 p_c, p_d)$ and $(p_c, T_1 p_d)$, where T_1 adds one second of ST and applies the quotient. Masses at most 10^{-9} are removed and each support is capped at its twelve largest actions. Let $k = \min(|D|, |C|)$; the first k actions in each support give

$$\begin{pmatrix} M_{D,C} & -\mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix} \begin{pmatrix} \sigma_{c,C} \\ u_c \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}, \quad \begin{pmatrix} M_{D,C}^\top & -\mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix} \begin{pmatrix} \sigma_{d,D} \\ u_d \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix}.$$

A pivot below 10^{-12} or mass below -10^{-12} rejects the candidate. The full-support attempt uses $D = C = \{1, \dots, 60\}$.

The residue is solved by the standard zero-sum pair

$$\begin{aligned} \max_{\sigma_d, v} \quad & v \quad \text{s.t.} \quad M^\top \sigma_d \geq v \mathbf{1}, \quad \mathbf{1}^\top \sigma_d = 1, \quad \sigma_d \geq 0, \\ \min_{\sigma_c, u} \quad & u \quad \text{s.t.} \quad M \sigma_c \leq u \mathbf{1}, \quad \mathbf{1}^\top \sigma_c = 1, \quad \sigma_c \geq 0. \end{aligned}$$

B Reproducibility

From the repository root, `uv run python -m dth complete` builds the checkpointed artifact with `src/dth/config/complete_full_v1.yaml`; its stored root midpoint is 0.0898500728095116.

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